Abdelrahman Mohamed Alnaqeeb

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Research interests: multi-agent guidance and control, mean-field games, flight control, GNC, optimization

Education

Zewail City of Science and Technology Expected June 2027
B.Sc. in Aerospace Engineering *GPA: 3.3/4.0 (to date)*
Relevant coursework: Control Systems; Advanced Control; Flight Dynamics & Control; Mechatronics; Numerical Analysis; Probability & Stochastic Processes; Gas Dynamics; Jet Propulsion; Fluid Mechanics II.

Research Experience

Graduation Thesis Sep 2026 – Jun 2027 (expected)
Mean-Field-Game Guidance of a Fixed-Wing Responder Fleet for Swarm Containment
Advisor: Dr. Ahmed Abdelsamea · Major coordinator: Dr. Ahmed Eltaweel
Status: formulation and experiment design drafted (scope revision v2, Oct 2026, under advisor review); solvers and simulations scheduled Nov 2026 – Apr 2027

- Formulated a team-versus-crowd mean-field game: a coupled HJB–Fokker–Planck system with containment compartments.
- Specified equilibrium computation by damped fictitious play, certifying each iterate by an exploitability residual.
- Specified a fidelity ladder from density planner to a 6-DOF fixed-wing UAV with a successive-loop-closure autopilot.
- Mapped each validity boundary to a classical prediction: loop sensitivity, describing function, Nyquist delay margin.
- Sketched a white-noise-limit bound $W_2 = O(\Pi_c^{1/2})$ linking OU gust correlation to planner diffusion (proof pending).
- Stated three falsifiable hypotheses with tolerances fixed in advance, a bootstrap protocol and a Latin-hypercube design.
- Set up a reproducible JAX/Diffrax environment and a Lean 4/Mathlib project in which solver invariants are to be proved.

Industry Research Internship

Lotus Power — CFD and Machine Learning Researcher Jun 2026 – present
Adjoint-based airflow optimization for a 20-tray industrial dryer *Supervisor: Dr. Basem Badr*

- Formulated tray-airflow uniformity as PDE-constrained topology optimization (Navier–Stokes/Brinkman model).
- Derived discrete-adjoint sensitivities through a density filter and projection; designed the finite-difference gradient checks.
- Designed the optimality-criteria update: multiplier bisection, move limits, trust-region backtracking, best-iterate restore.
- Cut tray-speed CV 0.495 $\rightarrow$ 0.09 (best iterate) and peak fan pressure 1115 $\rightarrow$ 544 Pa (2D reduced model; not 3D-validated).
- Directed an AI-assisted Python implementation (MAC-grid Newton solver, pytest suite); vectorization gave a 14.5$\times$ speed-up.
- Set up a 3D CFD pipeline (build123d, PyFluent, k–ω SST, 7.2M cells; not yet converged) and 1D fan-network sizing.

Selected Projects

CubeSat Attitude Determination & Control Simulation Feb 2026
Individual project; MATLAB/Simulink

- Generated a 1U CubeSat ADCS digital twin via the Simulink API: quaternion kinematics, wheel-momentum dynamics.
- Implemented B-dot detumbling, quaternion-error PD nadir tracking with torque limits, magnetorquer desaturation.
- Diagnosed a stalled detumble by time-constant analysis ($\sim 10^{10}$ s), retuned the B-dot gain, added 3D animation.

Active Quarter-Car Suspension: Pole Placement and LQR Jan 2026
Advanced Control course project, 4-person team; MATLAB/Simulink/Python

- Team state-space model and pole-placement baseline held deflection to 0.096 m with 0.53 s settling on a 0.1 m step.
- My part: discrete-time LQR Simulink model, switchable-terrain MATLAB GUI, Python RMS ride-comfort comparison.

3-DOF Robotic Arm: Kinematics, Digital Twin and Servo Control Dec 2025
Mechatronics course project; led all technical work in a 4-person team; MATLAB, Arduino, SolidWorks

- Derived forward/inverse kinematics; built a MATLAB digital twin with reach/joint limits commanding Arduino servos.
- Designed laser-cut Warren-truss plywood links and sized SG90 servos by static torque (safety factor 4.6).

Flight Dynamics & Control: Boeing 747 Attitude Autopilots May 2026
Course project; led all technical work in a 5-person team; MATLAB

- Compiled B747 stability derivatives; tuned pitch-attitude (PID) and roll-attitude (PI) hold loops for phugoid/spiral modes.

Optimization: Topology Optimization and Trajectory Design 2025 – 2026
Self-directed and course projects

- Satellite bracket (self-directed): compliance-minimizing topology optimization, mass $\leq 45\%$, stress cap, 37 iterations.
- Cold plate (Heat Transfer course): derived adjoint/KKT conditions for fluid–thermal channel topology optimization.
- JUICE trajectory: SPICE-anchored MGA-1DSM model, universal-variable Lambert solver, SQP-optimized DSM timing.

Propulsion Design: Ramjet and Cold-Gas Rocket Jan 2026
Course projects, 2-person team; MATLAB

- Ramjet at Mach 2.75: grid-searched two-ramp intake angles under shock and temperature limits for 6.26 kN net thrust.
- Cold-gas rocket: sized a CD nozzle and time-marched an adiabatic tank blowdown coupled to the choked nozzle.

Skills

Mathematics	Linear algebra, ODEs/PDEs, probability & stochastic processes, numerical analysis, optimization
Methods	State-space design, root locus, PID, pole placement, LQR; quaternion attitude dynamics; mean-field games (HJB–FPK); discrete adjoints, topology optimization; Newton continuation; Lambert targeting, SQP; RANS CFD
Software	MATLAB/Simulink (incl. programmatic model API), Python (NumPy, SciPy, JAX), ANSYS Fluent/PyFluent, OpenFOAM, COMSOL, SolidWorks Simulation, SPICE, Arduino C++, C++, Git, LaTeX